

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.
11. Norman S. Nise, *Control Systems Engineering*, 7th edition, Wiley, 2015.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 39****Topics covered :**

1. Construction of Fundamental set of solutions

- Consider the homogeneous linear system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \quad (1)$$

where $\mathbf{A}$ is a real-valued $n \times n$ matrix that has complex eigenvalues. Let $\lambda = p + jq$ be a complex eigenvalue of $\mathbf{A}$, with a corresponding complex eigenvector

$$\mathbf{v} = \mathbf{a} + j\mathbf{b}, \quad \text{where } \mathbf{a}, \mathbf{b} \in \mathbb{R}^n. \quad (2)$$

Since $\mathbf{A}$ is real, the complex conjugate $\lambda^* = p - jq$ is also an eigenvalue, and its corresponding eigenvector is $\mathbf{v}^* = \mathbf{a} - j\mathbf{b}$. Each eigenpair generates a complex-valued solution of the system

$$\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t}, \quad (3)$$

$$\mathbf{x}_2(t) = \mathbf{v}^*e^{\bar{\lambda}t}. \quad (4)$$

While these are valid solutions, they are generally complex-valued. Since the system matrix $\mathbf{A}$ is real and we are often interested in real-valued physical solutions, we must find a way to extract real-valued solutions from these complex expressions.

- Consider the complex-valued solution formed using the eigenvalue–eigenvector pair $\lambda = p + jq$, $\mathbf{v} = \mathbf{a} + j\mathbf{b}$

$$\mathbf{y}(t) = \mathbf{v}e^{\lambda t} = (\mathbf{a} + j\mathbf{b})e^{(p+jq)t}. \quad (5)$$

Equation (5) can be written as

$$\begin{aligned} \mathbf{y}(t) &= (\mathbf{a} + j\mathbf{b}) \cdot e^{pt}(\cos qt + j \sin qt) \\ &= e^{pt} [(\mathbf{a} \cos qt - \mathbf{b} \sin qt) + j(\mathbf{a} \sin qt + \mathbf{b} \cos qt)]. \end{aligned} \quad (6)$$

- Define the real and imaginary parts of $\mathbf{y}(t)$ as:

$$\mathbf{x}_1(t) = \text{Re}\{\mathbf{y}(t)\} = e^{pt} (\mathbf{a} \cos qt - \mathbf{b} \sin qt), \quad (7)$$

$$\mathbf{x}_2(t) = \text{Im}\{\mathbf{y}(t)\} = e^{pt} (\mathbf{a} \sin qt + \mathbf{b} \cos qt). \quad (8)$$

These functions are real-valued and are natural candidates for the fundamental set corresponding to the complex conjugate eigenvalue pair.

- Verification that $\mathbf{x}_1(t)$ satisfies the differential equation:¹ Recall that $\mathbf{y}(t)$ satisfies the system $\frac{d\mathbf{y}}{dt} = \mathbf{A}\mathbf{y}$. Since $\mathbf{A}$ is a real matrix, and differentiation is a linear operator that preserves real and imaginary parts independently, it follows that the real part $\mathbf{x}_1(t) = \text{Re}\{\mathbf{y}(t)\}$ must also satisfy:

$$\frac{d\mathbf{x}_1}{dt} = \mathbf{A}\mathbf{x}_1. \quad (9)$$

A similar argument shows that $\mathbf{x}_2(t)$ also satisfies the differential equation $\frac{d\mathbf{x}_2}{dt} = \mathbf{A}\mathbf{x}_2$.

- Linear independence of $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$: Suppose for contradiction that they are linearly dependent. Then there exists a scalar $c \in \mathbb{R}$ such that $\mathbf{x}_2(t) = c\mathbf{x}_1(t)$ for all t . This would imply that the imaginary part of $\mathbf{y}(t)$ is a constant multiple of its real part, which contradicts the fact that $\mathbf{a}$ and $\mathbf{b}$ are linearly independent vectors (since they form the real and imaginary parts of an eigenvector $\mathbf{v}$ corresponding to a non-real eigenvalue). Therefore, $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are linearly independent.
- Hence, $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ form a real-valued fundamental set of solutions corresponding to the complex conjugate eigenvalue pair $\lambda = p + jq$, $\lambda^* = p - jq$.

$$\mathbf{x}_1(t) = e^{pt} (\mathbf{a} \cos qt - \mathbf{b} \sin qt), \quad (10)$$

$$\mathbf{x}_2(t) = e^{pt} (\mathbf{a} \sin qt + \mathbf{b} \cos qt). \quad (11)$$

- The general real-valued solution to the system corresponding to the complex conjugate eigenvalues $\lambda = p + jq$ and $\lambda^* = p - jq$ is given by a linear combination of these two solutions:

$$\mathbf{x}(t) = c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t), \quad (12)$$

where c_1 and c_2 are real constants determined by the initial condition. Explicitly, this can be written as:

$$\mathbf{x}(t) = e^{pt} [c_1 (\mathbf{a} \cos qt - \mathbf{b} \sin qt) + c_2 (\mathbf{a} \sin qt + \mathbf{b} \cos qt)]. \quad (13)$$

- This representation makes it clear that the solution consists of oscillatory components (sine and cosine) modulated by the exponential factor e^{pt} . The sign of p determines the stability behavior:
 - If $p < 0$, the solution decays to zero — the system is stable and spirals into the origin.
 - If $p > 0$, the solution grows without bound — the system is unstable and spirals outward.
 - If $p = 0$, the solution is purely oscillatory — the system exhibits neutral stability (center).
- Proof of the result $\mathbf{A}\mathbf{a} = p\mathbf{a} - q\mathbf{b}$ and $\mathbf{A}\mathbf{b} = p\mathbf{b} + q\mathbf{a}$ used in the class: Let $\mathbf{v} = \mathbf{a} + j\mathbf{b}$ be a complex eigenvector of the real matrix $\mathbf{A}$, corresponding to the eigenvalue $\lambda = p + jq$. Then,

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}. \quad (14)$$

¹In class, we covered a different proof to indicate more interesting properties of the real and imaginary part of eigenvectors

Substituting $\mathbf{v} = \mathbf{a} + j\mathbf{b}$ and $\lambda = p + jq$, we have:

$$\mathbf{A}(\mathbf{a} + j\mathbf{b}) = (p + jq)(\mathbf{a} + j\mathbf{b}). \quad (15)$$

Equating real and imaginary parts of both sides results in

$$\text{Real part: } \mathbf{A}\mathbf{a} = p\mathbf{a} - q\mathbf{b}, \quad (16)$$

$$\text{Imaginary part: } \mathbf{A}\mathbf{b} = p\mathbf{b} + q\mathbf{a}. \quad (17)$$

Hence, the action of $\mathbf{A}$ on the real and imaginary parts of the complex eigenvector $\mathbf{v}$ results in real-valued linear combinations involving both $\mathbf{a}$ and $\mathbf{b}$, which reflects the rotational dynamics embedded in complex eigenstructure.